



Specifying and Analyzing Transactional Consistency Models with Predicates

Hengfeng Wei Si Liu Yuxing Chen

Hunan University

Texas A&M University

Renmin University of China

September 4, 2026

A semantic and graph-theoretic foundation for predicate operations

Item Operations vs. Predicate Operations

Read(x, v)

Write(x, v)

point access

returns one value

Employee

EID	Salary
0	80
1	100
2	60
3	40

PredRead(P, M)

$P : \text{Salary} < 80$

condition-based access

returns a set M of **matched** values

What we know about transactional consistency models
without predicates

Review: Intuitive or Operational Specs

Eswaran et al., 1976

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,
R.A. Lorie, and I.L. Traiger
IBM Research Laboratory
San Jose, California

SER

serial execution

Berenson et al., 1995

A Critique of ANSI SQL Isolation Levels

Hal Berenson	Microsoft Corp.	haroldb@microsoft.com
Phil Bernstein	Microsoft Corp.	philbe@microsoft.com
Jim Gray	U.C. Berkeley	gray@crl.com
Jim Melton	Sybase Corp.	jim.melton@sybase.com
Elizabeth O'Neil	UMass/Boston	conell@cs.umb.edu
Patrick O'Neil	UMass/Boston	poneil@cs.umb.edu

4.2 Snapshot Isolation

A transaction executing with *Snapshot Isolation* always reads data from a snapshot of the (committed) data as of the time the transaction started, called its *Start-Timestamp*. This time may be any time before the transaction's first Read. A transaction running in Snapshot Isolation is never blocked attempting a read as long as the snapshot data from its *Start-Timestamp* can be maintained. The transaction's writes (updates, inserts, and deletes) will also be reflected in this snapshot, to be read again if the transaction accesses (i.e., reads or updates) the data a second time. Updates by other transactions active after the transaction *Start-Timestamp* are invisible to the transaction.

Snapshot Isolation is a type of multiversion concurrency control. It extends the Multiversion Mixed Method described in [BHG], which allowed snapshot reads by read-only transactions.

When the transaction T1 is ready to commit, it gets a *Commit-Timestamp*, which is larger than any existing *Start-Timestamp* or *Commit-Timestamp*. The transaction successfully commits only if no other transaction T2 with a *Commit-Timestamp* in T1's interval [*Start-Timestamp*, *Commit-Timestamp*] wrote data that T1 also wrote. Otherwise, T1 will abort. This feature, called *First-committer-wins* prevents lost updates (phenomenon P4). When T1 commits, its changes become visible to all transactions whose *Start-Timestamps* are larger than T1's *Commit-Timestamp*.

SI

snapshot + write-conflict control

Review: Axiomatic Specs

Cerone, Bernardi, and Gotsman, 2015

A Framework for Transactional Consistency Models with Atomic Visibility

Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman

IMDEA Software Institute, Madrid, Spain

(VIS, AR)

Crooks et al., 2017

Seeing is Believing: A Client-Centric Specification of Database Isolation

Natacha Crooks

The University of Texas at Austin and Cornell University

Lorenzo Alvisi

The University of Texas at Austin and Cornell University

Yuer Pu

Cornell University

Allen Clement

Google, Inc.

Client-Centric Views

Complexity:OOPSLA2019

On the Complexity of Checking Transactional Consistency

RANADEEP BISWAS, Université de Paris, IRIF, CNRS, France

CONSTANTIN ENEA, Université de Paris, IRIF, CNRS, France

Checking Complexity

Review: Dependency Graphs

Eswaran et al., 1976

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,
R.A. Lorie, and I.L. Traiger
IBM Research Laboratory
San Jose, California

CC:STOC1976

CONCURRENCY CONTROL FOR DATABASE SYSTEMS

R.E. Stearns, P.M. Lewis II and D.J. Rosenkrantz

General Electric Research & Development Center
Schenectady, N. Y. 12345

EXTENDED ABSTRACT

Adya, 1999

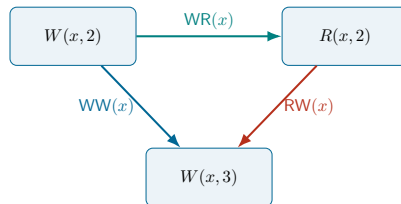
Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy



Review: SER Characterization Theorem

CC:STOC1976

CONCURRENCY CONTROL FOR DATABASE SYSTEMS

R.E. Stearns, P.M. Lewis II and D.J. Rosenkrantz
General Electric Research & Development Center
Schenectady, N. Y. 12345

EXTENDED ABSTRACT

**Weak Consistency: A Generalized Theory and Optimistic
Implementations for Distributed Transactions**

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

Adya, 1999

Theorem

*An item-operation history is serializable iff its
dependency graph is internally consistent and acyclic.*

Review: SI Characterization Theorem

4.2 Snapshot Isolation

A transaction executing with *Snapshot Isolation* always reads data from a snapshot of the (committed) data as of the time the transaction started, called its *Start-Timestamp*. This time may be any time before the transaction's first Read. A transaction running in Snapshot Isolation is never blocked attempting a read as long as the snapshot data from its *Start-Timestamp* can be maintained. The transaction's writes (updates, inserts, and deletes) will also be reflected in this snapshot, to be read again if the transaction accesses (i.e., reads or updates) the data a second time. Updates by other transactions active after the transaction *Start-Timestamp* are invisible to the transaction.

Snapshot Isolation is a type of multiversion concurrency control. It extends the Multiversion Mixed Method described in [BHG], which allowed snapshot reads by read-only transactions.

When the transaction T1 is ready to commit, it gets a *Commit-Timestamp*, which is larger than any existing *Start-Timestamp* or *Commit-Timestamp*. The transaction successfully commits only if no other transaction T2 with a *Commit-Timestamp* in T1's interval [*Start-Timestamp*, *Commit-Timestamp*] wrote data that T1 also wrote. Otherwise, T1 will abort. This feature, called *First-committer-wins* prevents lost updates (phenomenon P4). When T1 commits, its changes become visible to all transactions whose *Start-Timestamps* are larger than T1's *Commit-Timestamp*.

Theorem

An item-operation history satisfies SI iff its dependency graph is internally consistent and every cycle has two adjacent RW edges.

This exact correspondence is the baseline that predicates must recover.

Review: Downstream Tasks

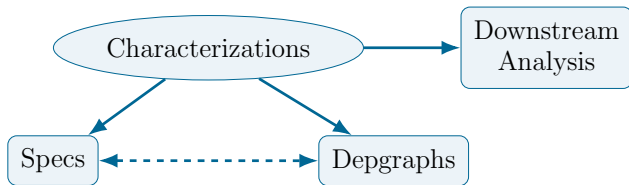
Task	Representative work
Black-box checking	Elle, COBRA, PolySI, VeriStrong
CC protocol verification	execution-based verification
Robustness	robustness against weaker consistency

All depend on a faithful link between observed histories and consistency.

Kingsbury and Alvaro, 2020; Tan et al., 2020; Huang et al., 2023; Cai et al., 2025; Bernardi and Gotsman, 2016

Review (Summary)

What we know about transactional consistency models
without predicates



What we know about transactional consistency models
with predicates

Review: Intuitive or Operational Specs

Eswaran et al., 1976

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,
R.A. Lorie, and I.L. Traiger
IBM Research Laboratory
San Jose, California

SER

serial execution

Berenson et al., 1995

A Critique of ANSI SQL Isolation Levels

Hal Berenson	Microsoft Corp.	haroldb@microsoft.com
Phil Bernstein	Microsoft Corp.	philbe@microsoft.com
Jim Gray	U.C. Berkeley	gray@cs.berkeley.edu
Jim Melton	Sybase Corp.	jim.melton@sybase.com
Elizabeth O'Neil	UMass/Boston	oneil@cs.umb.edu
Patrick O'Neil	UMass/Boston	poneil@cs.umb.edu

4.2 Snapshot Isolation

A transaction executing with *Snapshot Isolation* always reads data from a snapshot of the (committed) data as of the time the transaction started, called its *Start-Timestamp*. This time may be any time before the transaction's first Read. A transaction running in Snapshot Isolation is never blocked attempting a read as long as the snapshot data from its *Start-Timestamp* can be maintained. The transaction's writes (updates, inserts, and deletes) will also be reflected in this snapshot, to be read again if the transaction accesses (i.e., reads or updates) the data a second time. Updates by other transactions active after the transaction *Start-Timestamp* are invisible to the transaction.

Snapshot Isolation is a type of multiversion concurrency control. It extends the Multiversion Mixed Method described in [BHG], which allowed snapshot reads by read-only transactions.

When the transaction T1 is ready to commit, it gets a *Commit-Timestamp*, which is larger than any existing *Start-Timestamp* or *Commit-Timestamp*. The transaction successfully commits only if no other transaction T2 with a *Commit-Timestamp* in T1's interval [*Start-Timestamp*, *Commit-Timestamp*] wrote data that T1 also wrote. Otherwise, T1 will abort. This feature, called *First-committer-wins* prevents lost updates (phenomenon P4). When T1 commits, its changes become visible to all transactions whose *Start-Timestamps* are larger than T1's *Commit-Timestamp*.

SI

snapshot + write-conflict control

Still applicable

Review: Axiomatic Specs

Bouajjani, Enea, and Román-Calvo, 2025

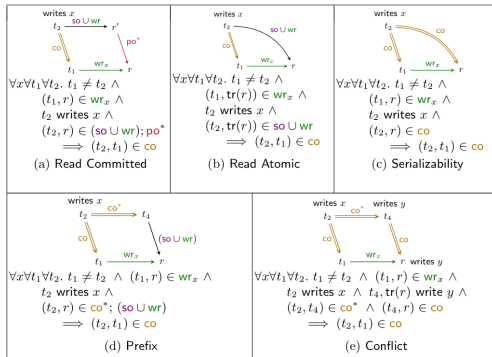
On the Complexity of Checking Mixed Isolation Levels for SQL Transactions

Ahmed Bouajjani¹, Constantin Enea², and Enrique Román-Calvo¹



¹ Université Paris Cité, CNRS, IRIF
{abou, calvo}@irif.fr

² LIX, École Polytechnique, CNRS and
Institut Polytechnique de Paris
cenea@lix.polytechnique.fr



checking-oriented consistency axioms

Review: Dependency Graphs

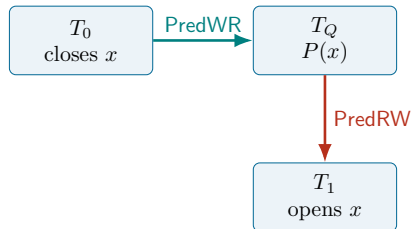
Adya, 1999

**Weak Consistency: A Generalized Theory and Optimistic
Implementations for Distributed Transactions**

by
Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy



A predicate read depends on a witness and
on later result-changing writers.

Review: Dependency Graphs

“Generalized Isolation Level
Definitions” 2000

Generalized Isolation Level Definitions

Atul Adya

Microsoft Research,
1 Microsoft Way,
Redmond, WA 98007
adya@microsoft.com

Barbara Liskov

Laboratory for Computer Science,
Massachusetts Inst. of Technology,
Cambridge, MA 02139
liskov@cs.mit.edu

Patrick O’Neil

Univ. of Massachusetts,
Boston, MA 02125-3393
poneil@cs.umb.edu

generalized isolation

Sun and Zou, 2025

Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

earliest result change

Bouajjani, Enea, and Román-Calvo,
2025

**On the Complexity of Checking Mixed Isolation
Levels for SQL Transactions**

Ahmed Bouajjani¹, Constantin Enea², and Enrique Román-Calvo¹



¹ Université Paris Cité, CNRS, IRIF
{abou, calvo}@irif.fr

² LIX, École Polytechnique, CNRS and
Institut Polytechnique de Paris
cenea@lix.polytechnique.fr

all later writers

Predicate-read dependency definitions differ.

Review: Dependency Graphs

Sun and Zou, 2025

Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

earliest result change

“Generalized Isolation Level
Definitions” 2000

Generalized Isolation Level Definitions

Atul Adya

Microsoft Research,
1 Microsoft Way,
Redmond, WA 98007
adya@microsoft.com

Barbara Liskov

Laboratory for Computer Science,
Massachusetts Inst. of Technology,
Cambridge, MA 02139
liskov@lcs.mit.edu

Patrick O’Neil

Univ. of Massachusetts,
Boston, MA 02125-3393
poneil@cs.umb.edu

generalized isolation

Bouajjani, Enea, and Román-Calvo,
2025

**On the Complexity of Checking Mixed Isolation
Levels for SQL Transactions**

Ahmed Bouajjani¹, Constantin Enea², and Enrique Román-Calvo¹



¹ Université Paris Cité, CNRS, IRIF
{abou, calvo}@irif.fr

² LIX, École Polytechnique, CNRS and
Institut Polytechnique de Paris
cenea@lix.polytechnique.fr

all later writers

$$\text{PredRW}^+ \subseteq \text{PredRW} \subseteq \text{PredRW}^-$$

predicate-anti-dependency definitions differ.

Review: SER Characterization

Predicate extensions use different edges, but lack a single rigorous semantic proof bridge.

3.4 Correctness and Flexibility of the New Specifications

Our conditions for PL-3 provide the well-known notion of conflict-serializability [BHG87, GR93]. That is, $DSG(H)$ is acyclic, and G1a, and G1b are satisfied for a history H iff H is conflict-serializable. We can prove the equivalence of our PL-3 conditions with conflict-serializability using the proof given in [GR93]; our DSGs are similar to their graphs. This equivalence can also be proved along the lines of the proof given in [BHG87] for view-serializability.

Bernstein, Hadzilacos, and Goodman, 1987; Gray and Reuter, 1993; Adya, 1999; “Generalized Isolation Level Definitions” 2000; Fekete et al., 2005; Clark, Donaldson, et al., 2024; Clark, Rigger, et al., 2026; Sun and Zou, 2025

Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions

by

Atul Adya

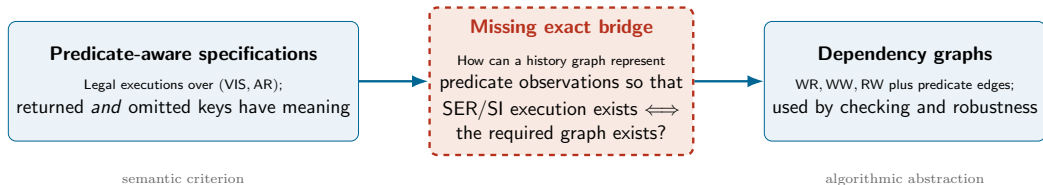
Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

Low level info: Which predicate edges
preserve this exact characterization?

Adya, 1999

The Missing Bridge

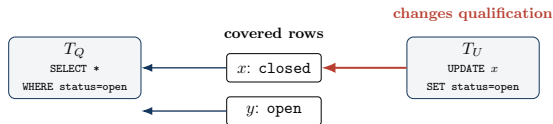


The gap

Predicate-aware **specifications** say which executions are legal; predicate **dependency graphs** are useful only when an *exact, two-way characterization* connects them. That bridge is established for item operations, but was missing for predicate observations.

Cerone and Gotsman, 2018; Bouajjani, Enea, and Román-Calvo, 2025; Adya, 1999; Sun and Zou, 2025

Why Predicates Make the Bridge Nontrivial



result: $\{y\}$

Key observation: $x \notin \text{result}$
still records an observation of x .

Item read: records the value of one named key.

Predicate read: observes a *set*: both returned keys and covered keys that are absent.

A concurrent write can therefore change the result by making an absent key match—the classical phantom.

A graph edge must carry enough information to explain that absence.

Berenson et al., 1995; Adya, 1999

What the Missing Bridge Must Deliver

Semantic side

A predicate observation is explained by the latest visible version of every externally observed key, including an omitted key.

Graph side

Predicate read- and anti-dependencies must encode the explanatory version and every observation-changing later write.

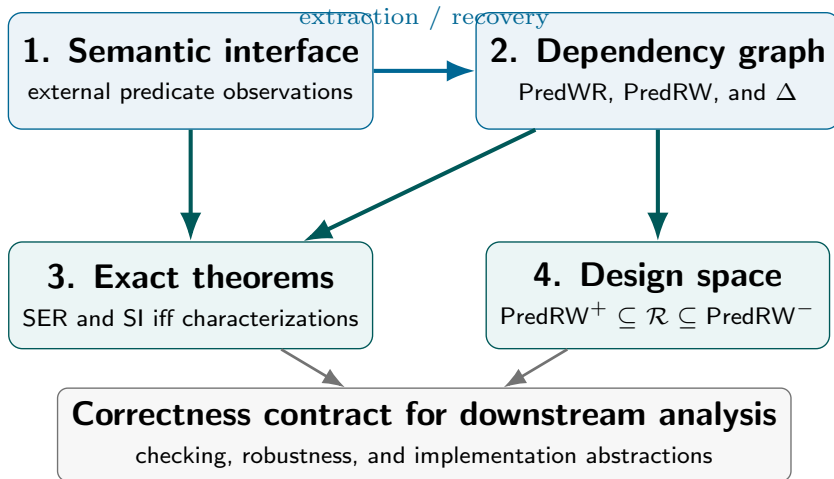
Characterization obligation

SER/SI execution exists $\iff$ a graph satisfying the corresponding criterion exists.

The reverse implication is the hard one: construct a legal execution *from* a graph without losing predicate semantics.

Cerone and Gotsman, 2018

Contributions: Build and Exploit the Bridge



Four Contributions, One Coherent Story

1. **Semantic interface.** Extend (VIS, AR) with item-wise external predicate observations, so both returned and omitted keys have a precise version explanation.
2. **Predicate dependency graphs.** Define PredWR for the explanatory writer and observation-sensitive PredRW using the change predicate Δ .
3. **Exact SER/SI characterizations.** Prove the same execution-graph equivalence style as the item-only results, while resolving the predicate-specific SI witness ambiguity.
4. **Proof-theoretic design space.** Show that every relation between a lower frontier and an upper structural closure preserves the same existential history-level characterization.

Cerone and Gotsman, 2018; Adya, 1999; Bouajjani, Enea, and Román-Calvo, 2025; Sun and Zou, 2025

Claim Boundary: Theory First, Engineering Enabled

We establish

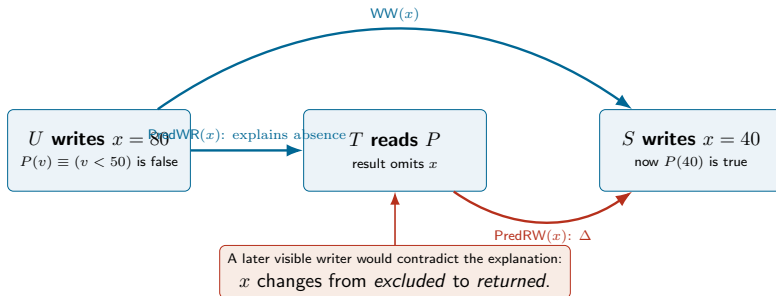
- a predicate-aware semantic-graph correspondence;
- exact graph criteria for SER and SI; and
- a sound interval of admissible predicate anti-dependencies.

We deliberately do not claim

- a new full SQL black-box checker or a cost improvement;
- that every DBMS lock event is literally a graph edge; or
- that the interval relations have identical fixed-graph edges.

The contribution is a reusable correctness contract for those downstream tools.

Technical Insight 1: Make Absence a First-Class Observation



External predicate consistency

For each externally observed key x , choose the AR-latest writer visible to the reader. Its value must equal the returned value when x appears in the result, and must fail the predicate when x is omitted.

Technical Insight 2: Witnesses Turn a Semantic Mismatch into an Edge

Graph witness

$U \xrightarrow{\text{PredWR}(x)} T$ states that U supplies the version by which T externally observes x .

Observation-changing successor

If $U \xrightarrow{\text{WW}(x)} S$ and replacing U 's value by S 's value changes membership or the returned value, then

$$T \xrightarrow{\text{PredRW}(x)} S.$$

Proof move

In the graph-to-execution direction, assume the constructed execution violates an external predicate observation. The graph witness and the execution's latest visible writer then disagree in an observable way, forcing PredRW and hence a forbidden order/cycle.

Predicate analogue of recovering read-from facts from a graph.

Technical Insight 3: Keep Exactly the Proof-Relevant Edges

$$\text{PredRW}^+ \subseteq \mathcal{R} \subseteq \text{PredRW}^-$$

Lower frontier

The earliest later write that changes the observation. It is enough to expose a predicate-consistency violation.

Baseline

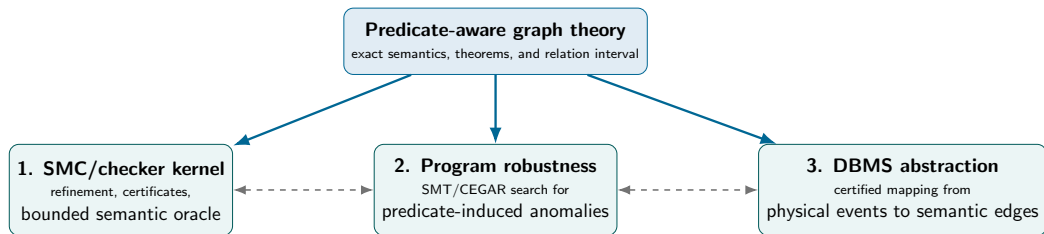
All observation-changing later writers. This is the direct “changes matches” reading.

Upper closure

All later writers sharing a PredWR/WW witness. It is safe to orient in a witnessing execution.

Different fixed graphs; the same existential SER/SI history characterization. The theorem separates correctness from representation.

Future Work: Downstream Analysis Tasks



Three directions share one prerequisite: preserve the semantic bridge while moving from histories to algorithms and implementations.

1. Predicate-Aware Stateless Model Checking

Theory-guided checker kernel

From normalized predicate histories, construct a common PredWR/WW base and use lower/baseline/upper relations for SER/SI checks. A lower persistent cycle can reject; an upper acyclic graph can accept; ambiguous cases are refined or solved.

Make it scientifically credible

- validate against exhaustive bounded (VIS, AR) search;
- emit a checkable witness/cycle certificate; and
- measure edge density, Δ tests, memory, and solver work.

The target is a relation-policy laboratory or SMC kernel—not merely another end-to-end black-box checker.

Bouajjani, Enea, and Román-Calvo, 2025; Sun and Zou, 2025

2–3. Program Robustness and Faithful Implementations

2. Predicate-aware robustness analysis

Lift item-only robustness questions from histories to transaction programs: do all SI executions of a program remain serializable when predicate reads can create phantoms? A finite predicate/write abstraction and the graph criterion can drive SMT/CEGAR search for a violating program path.

Success: a sound abstraction theorem and counterexample traces that explain the decisive predicate edge.

3. Certified mapping from DBMS events

Relate recorded predicate/range-conflict events to semantic keys, writer order, coverage, and Δ . Prove when a mechanism is a sound lower or upper approximation, rather than labeling physical locks as semantic edges by fiat.

Success: one DBMS adapter with a stated abstraction theorem and certificates validated against captured histories.

Fekete et al., 2005; Bernardi and Gotsman, 2016; Cerone and Gotsman, 2018